

Use the following information to answer the next question.

A student designed a calorimetry experiment to determine the mass of water that could be heated by 400 kJ of energy released by burning butane gas, and recorded the following results.

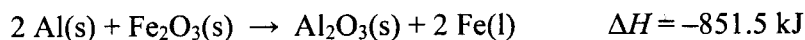
Initial temperature of water	21.6 °C
Final temperature of water	55.6 °C
Energy released	400 kJ

Numerical Response

1. The mass of water involved in this calorimetry experiment is _____ kg.

Use the following information to answer the next question.

A thermite reaction is a highly exothermic process that is used in welding massive steel objects such as ship propellers and train rails. The reaction can be represented by the following equation.



Numerical Response

2. If the heat produced by the reaction of 1.00 mol of iron(III) oxide were absorbed by 7.40 kg of $\text{H}_2\text{O(l)}$ at room temperature, then the resulting temperature change of the water would be _____ °C.
3. The amount of heat released by a substance when the temperature of 250.0 g of water increases from 20.0 °C to 35.0 °C is
- A. 12.4 kJ
B. 13.7 kJ
C. 15.7 kJ
D. 57.6 kJ

4. Which of the following reactions would release the **largest** amount of energy?

- A. $2 \text{Na(s)} + \text{Cl}_2\text{(g)} \rightarrow 2 \text{NaCl(s)}$
- B. $\text{CaO(s)} + 3 \text{C(s)} + 462.3 \text{ kJ} \rightarrow \text{CaC}_2\text{(s)} + \text{CO(g)}$
- C. $\text{P}_4\text{(s)} + 10 \text{Cl}_2\text{(g)} \rightarrow 4 \text{PCl}_5\text{(g)} \quad \Delta H^\circ = -1\,772 \text{ kJ}$
- D. $\text{C}_2\text{H}_4\text{(g)} + 3 \text{O}_2\text{(g)} \rightarrow 2 \text{CO}_2\text{(g)} + 2 \text{H}_2\text{O(g)} + 1\,322.9 \text{ kJ}$

5. The energy required to completely decompose 1.50 mol of $\text{PCl}_3\text{(l)}$ is

- A. 480 kJ
- B. 320 kJ
- C. 287 kJ
- D. 213 kJ

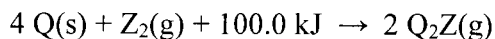
Use the following information to answer the next question.

A fire fuelled by burning magnesium metal produces a brilliant white light and intense heat, which makes fighting the fire very difficult. Firefighters extinguish this type of fire by smothering it with sand.

6. Which of the following equations represents the burning of magnesium?

- A. $\text{Mg(s)} + \frac{1}{2} \text{O}_2\text{(g)} + 601.6 \text{ kJ} \rightarrow \text{MgO(s)}$
- B. $\text{Mg(s)} + \frac{1}{2} \text{O}_2\text{(g)} \rightarrow \text{MgO(s)} + 601.6 \text{ kJ}$
- C. $\text{Mg(s)} + \frac{1}{2} \text{O}_2\text{(g)} \rightarrow \text{MgO(s)} \quad \Delta H^\circ = +601.6 \text{ kJ}$
- D. $2 \text{Mg(s)} + \text{O}_2\text{(g)} + 1\,203.2 \text{ kJ} \rightarrow 2 \text{MgO(s)}$

Use the following information to answer the next question.



7. The ΔH value for the reverse reaction would be

- A. -100.0 kJ
- B. -50.0 kJ
- C. $+50.0 \text{ kJ}$
- D. $+100.0 \text{ kJ}$

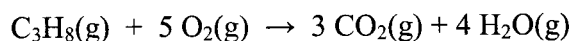
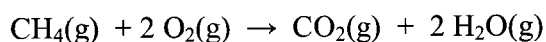
Use the following information to answer the next question.

Fuel	Molar Enthalpy of Combustion
Methane	−802.5 kJ/mol
Ethane	−1 428.4 kJ/mol
Butane	−2 657.3 kJ/mol
Octane	−5 074.1 kJ/mol

8. An interpretation that applies to the data in the table above is that the greater the number of carbon atoms in a fossil fuel molecule, the
- A. lower the amount of energy stored in the molecule
 - B. greater the amount of energy released during combustion
 - C. greater the strength of each carbon-carbon covalent bond
 - D. lower the chemical amount of oxygen required for complete combustion

Use the following information to answer the next question.

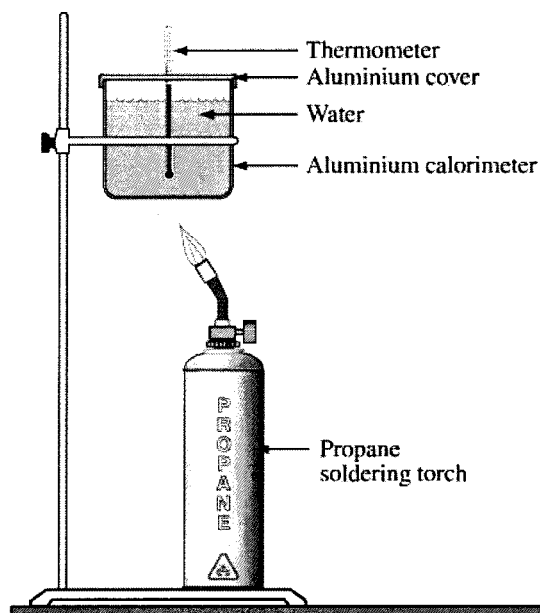
In an experiment, technicians compared the use of methane and propane as fuels. They burned samples of the two fuels and used a calorimeter to determine the energy transferred. The combustion reactions are represented by the following equations.



9. Which of the following statements correctly classifies two of the variables in the technicians' experiment?
- A. The manipulated variable is the type of fuel, and a controlled variable is the type of calorimeter.
 - B. The responding variable is the final temperature of the water, and a controlled variable is the type of fuel.
 - C. The manipulated variable is the type of fuel, and a responding variable is the temperature change of the fuel.
 - D. The responding variable is the final temperature of the water, and a controlled variable is the temperature change of the fuel.

Use the following information to answer the next question.

A technician performed an experiment to determine the molar enthalpy of combustion of propane in a soldering torch, as represented by the diagram below.



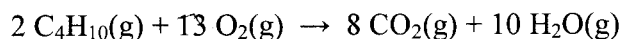
10. If the experimental value of the molar enthalpy of combustion of propane in the technician's calorimetry experiment is significantly different from the theoretical value, then the technician could reduce the discrepancy in the data by
- A. using a glass beaker to hold the water
 - B. creating an enclosing shield around the apparatus
 - C. raising the aluminium calorimeter higher above the flame
 - D. decreasing the mass of water in the aluminium calorimeter
11. If 44.7 kJ of energy is released when 1.65 g of ethanol, $\text{C}_2\text{H}_5\text{OH}(\text{l})$, is burned in a calorimeter, then the molar enthalpy of combustion of ethanol is
- A. -1.67 kJ/mol
 - B. -27.1 kJ/mol
 - C. -73.8 kJ/mol
 - D. $-1.25 \times 10^3 \text{ kJ/mol}$

Numerical Response

12. The energy released when 0.500 mol of AgI(s) is formed from its elements is _____ kJ.
13. Which of the following equations represents the balanced chemical equation for the decomposition of water and the energy transfer during the reaction?
- A. $\text{H}_2(\text{g}) + \frac{1}{2} \text{O}_2(\text{g}) \rightarrow \text{H}_2\text{O}(\text{l}) + 285.8 \text{ kJ}$
- B. $\text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_2(\text{g}) + \frac{1}{2} \text{O}_2(\text{g}) + 285.8 \text{ kJ}$
- C. $2 \text{H}_2\text{O}(\text{l}) + 285.8 \text{ kJ} \rightarrow 2 \text{H}_2(\text{g}) + \text{O}_2(\text{g})$
- D. $\text{H}_2\text{O}(\text{l}) + 285.8 \text{ kJ} \rightarrow \text{H}_2(\text{g}) + \frac{1}{2} \text{O}_2(\text{g})$
14. When used as a fuel in automobiles, ethanol burns more efficiently than fossil fuels. The balanced equation, including the appropriate enthalpy change for the combustion of ethanol, is
- A. $\text{C}_2\text{H}_5\text{OH}(\text{l}) + 3 \text{O}_2(\text{g}) \rightarrow 2 \text{CO}_2(\text{g}) + 3 \text{H}_2\text{O}(\text{g}) \quad \Delta H^\circ = -1\,234.8 \text{ kJ}$
- B. $\text{C}_2\text{H}_5\text{OH}(\text{l}) + 3 \text{O}_2(\text{g}) \rightarrow 2 \text{CO}_2(\text{g}) + 3 \text{H}_2\text{O}(\text{g}) \quad \Delta H^\circ = +1\,234.8 \text{ kJ}$
- C. $\text{C}_2\text{H}_5\text{OH}(\text{l}) \rightarrow 2 \text{C}(\text{s}) + 3 \text{H}_2(\text{g}) + \frac{1}{2} \text{O}_2(\text{g}) \quad \Delta H^\circ = +277.6 \text{ kJ}$
- D. $\text{C}_2\text{H}_5\text{OH}(\text{l}) \rightarrow 2 \text{C}(\text{s}) + 3 \text{H}_2(\text{g}) + \frac{1}{2} \text{O}_2(\text{g}) \quad \Delta H^\circ = -277.6 \text{ kJ}$

Use the following information to answer the next question.

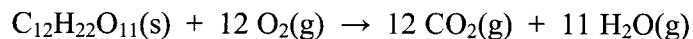
Disposable lighters contain butane gas which undergoes combustion, as represented by the following equation.



15. According to the equation above, the molar enthalpy change for the combustion of butane gas is
- A. $-2\,657.3 \text{ kJ/mol}$
- B. $-2\,877.3 \text{ kJ/mol}$
- C. $-5\,314.6 \text{ kJ/mol}$
- D. $-5\,754.6 \text{ kJ/mol}$

Use the following information to answer the next **two** questions.

Sucrose, $\text{C}_{12}\text{H}_{22}\text{O}_{11}(\text{s})$, is commonly used to sweeten food. In order to determine the molar enthalpy of combustion for sucrose, a technician burns a 0.0150 mol sample of sucrose using a copper calorimeter that contains 250.0 g of water. The combustion of sucrose is represented by the following equation.



16. If the temperature change of the water in the calorimeter is $+55.5\text{ }^{\circ}\text{C}$, then the experimental molar enthalpy of combustion for sucrose is approximately
- A. $-8.72 \times 10^{-1} \text{ kJ/mol}$
B. $-1.40 \times 10^1 \text{ kJ/mol}$
C. $-5.81 \times 10^1 \text{ kJ/mol}$
D. $-3.88 \times 10^3 \text{ kJ/mol}$
17. If included as a term in the equation above, energy is a *i* , and the water in the calorimeter undergoes a change primarily in *ii* energy.

The statement above is completed by the information in row

Row	<i>i</i>	<i>ii</i>
A.	reactant	kinetic
B.	reactant	potential
C.	product	kinetic
D.	product	potential

Use the following information to answer the next question.

A student uses a calorimeter to determine the molar enthalpy of solution for solid ammonium nitrate. The student assumes that the calorimeter neither gains nor loses heat during the experiment and that the density and specific heat capacity for the final solution are the same as those of water. The data was collected and recorded in the following table.

Mass of calorimeter	25.45 g
Mass of calorimeter and water	175.45 g
Mass of ammonium nitrate	1.68 g
Initial temperature of water	22.30 °C
Final temperature of water	20.98 °C

Numerical Response

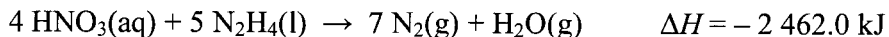
18. The experimental molar enthalpy of solution of ammonium nitrate is $\pm$ _____ kJ/mol.

Use the following information to answer the next question.

In a calorimetry experiment, a student found that the theoretical value for the molar enthalpy of combustion for methanol was higher than the value that she obtained in the laboratory.

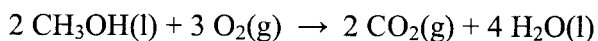
19. Which of the following explanations could account for the discrepancy between the experimental value and the theoretical value for the molar enthalpy of combustion for methanol?
- A. The mass of methanol was lower than measured
 - B. The final temperature of the calorimeter contents was higher than measured
 - C. The initial temperature of the calorimeter contents was lower than measured
 - D. The incomplete combustion of methanol produced a mixture of CO(g) and CO₂(g) in the calorimeter

Use the following information to answer the next question.



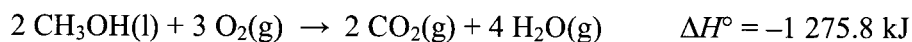
20. Which statement is correct for the indicated reaction?
- A. 205.2 kJ are absorbed per mole of $\text{H}_2\text{O}(\text{g})$ produced
 - B. 351.7 kJ are absorbed per mole of $\text{N}_2(\text{g})$ produced
 - C. 615.5 kJ are released per mole of $\text{HNO}_3(\text{aq})$ consumed
 - D. 492.4 kJ are released per mole of $\text{N}_2\text{H}_4(\text{g})$ produced

Use the following information to answer the next question.



21. The amount of thermal energy released for the complete combustion of one mole of methanol is
- A. 725.9 kJ/mol
 - B. 563.4 kJ/mol
 - C. 277.6 kJ/mol
 - D. 223.6 kJ/mol

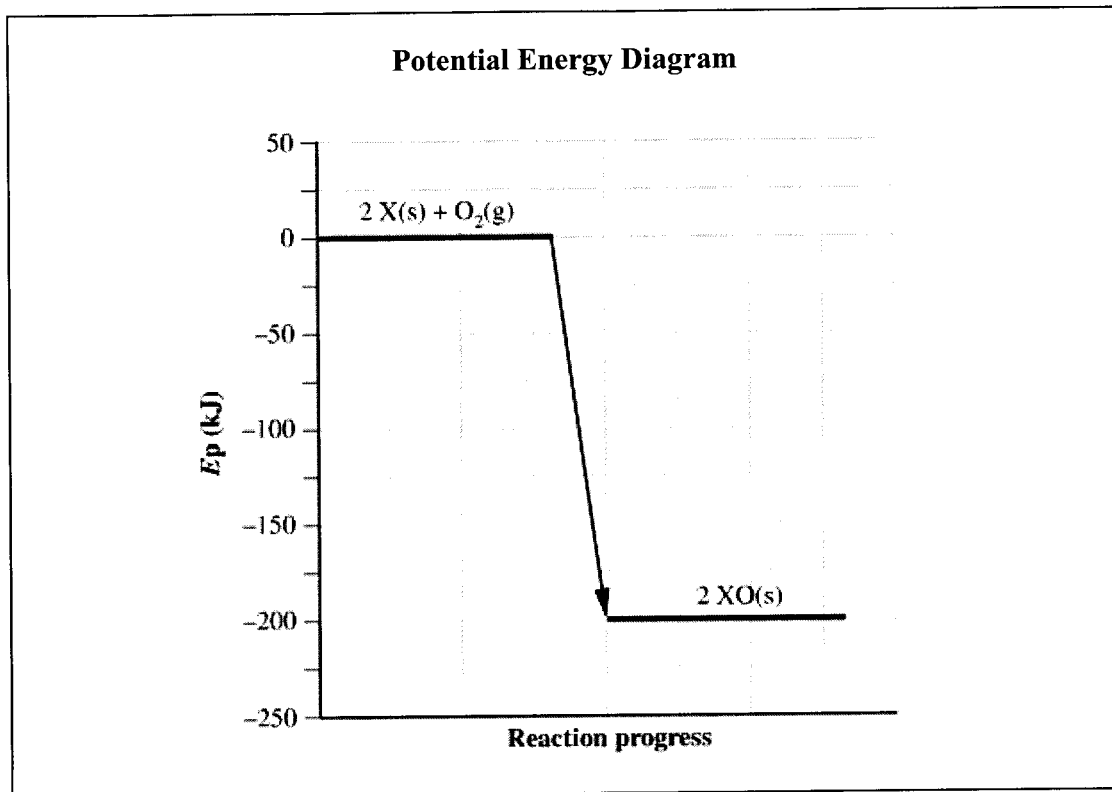
Use the following information to answer the next **two** questions.



Numerical Response

22. The mass of methanol that burns to produce an enthalpy change of $-9.00 \times 10^4 \text{ kJ}$ is _____ kg.
23. The amount of methanol that must be burned to raise the temperature of 250.0 g of water from 20.0°C to 35.0°C is
- A. $6.16 \times 10^{-3} \text{ mol}$
 - B. $1.23 \times 10^{-2} \text{ mol}$
 - C. $2.46 \times 10^{-2} \text{ mol}$
 - D. $2.46 \times 10^1 \text{ mol}$

Use the following information to answer the next question.



24. The reaction represented by the graph above is i and the energy term would be included as a ii in the balanced chemical equation.

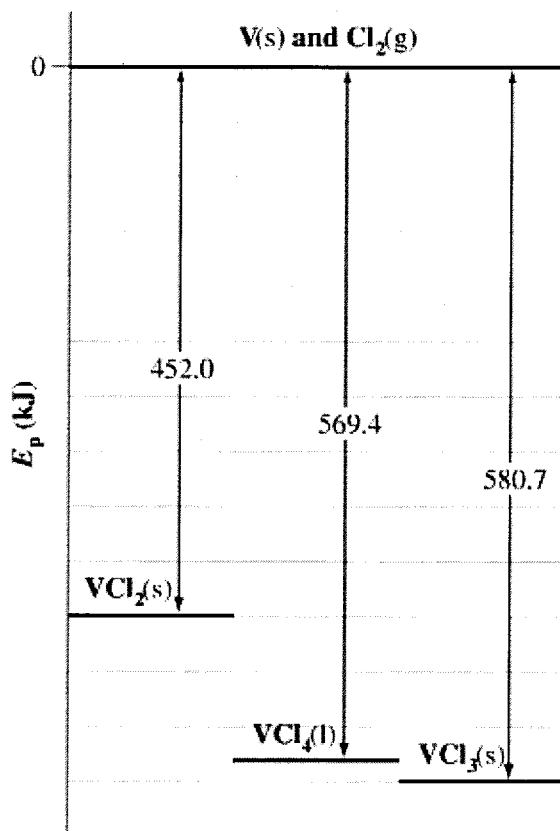
The statement above is completed by the information in row

Row	i	ii
A.	exothermic	product
B.	exothermic	reactant
C.	endothermic	product
D.	endothermic	reactant

Use the following information to answer the next question.

The following diagram illustrates the formation enthalpies of V(s), Cl₂(g), and a selection of their compounds.

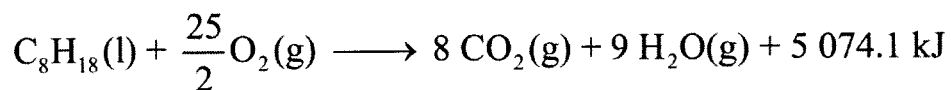
Formation Enthalpies of Vanadium Chlorides



Numerical Response

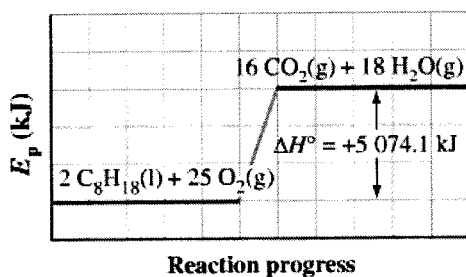
25. The amount of energy absorbed when 0.350 mol of $VCl_4(l)$ decomposes to form $VCl_2(s)$ and $Cl_2(g)$ is _____ kJ.

Use the following information to answer the next question.

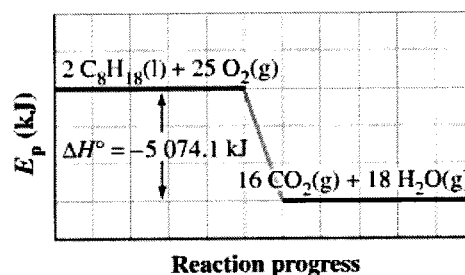


26. Which of the following potential energy diagrams represents the equation above?

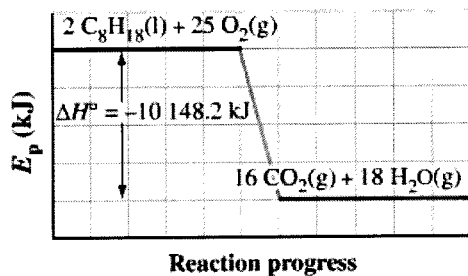
A.



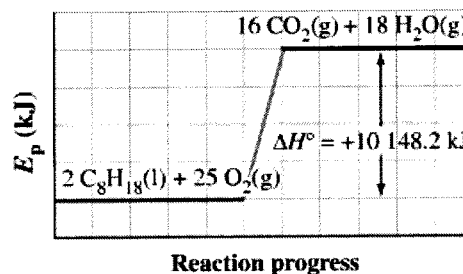
B.



C.



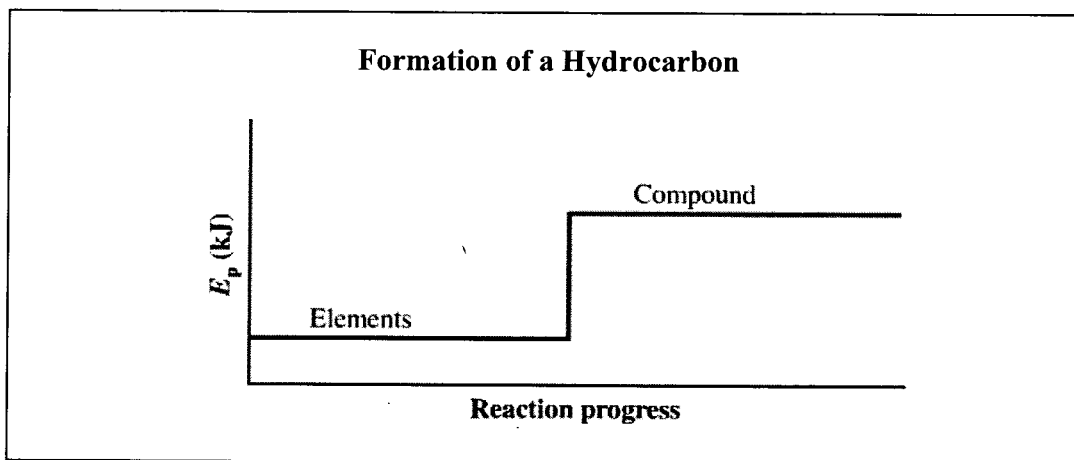
D.



27. One reason that could explain why dynamite releases a large quantity of energy when it explodes is that the reaction is

- A. exothermic, and the products have more potential energy than the reactants
- B. exothermic, and the reactants have more potential energy than the products
- C. endothermic, and the reactants have more potential energy than the products
- D. endothermic, and the products have more potential energy than the reactants

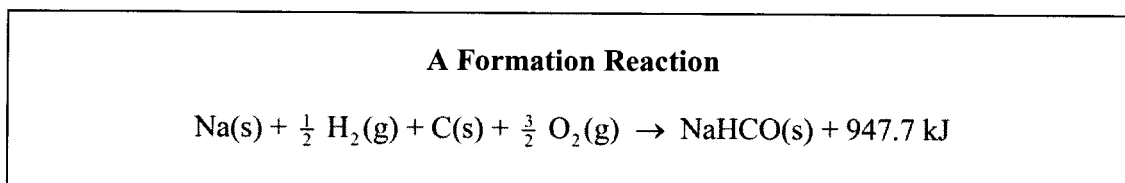
Use the following information to answer the next question.



28. The graph above could represent the formation of

- A. ethane
- B. ethene
- C. butane
- D. pentane

Use the following information to answer the next question.

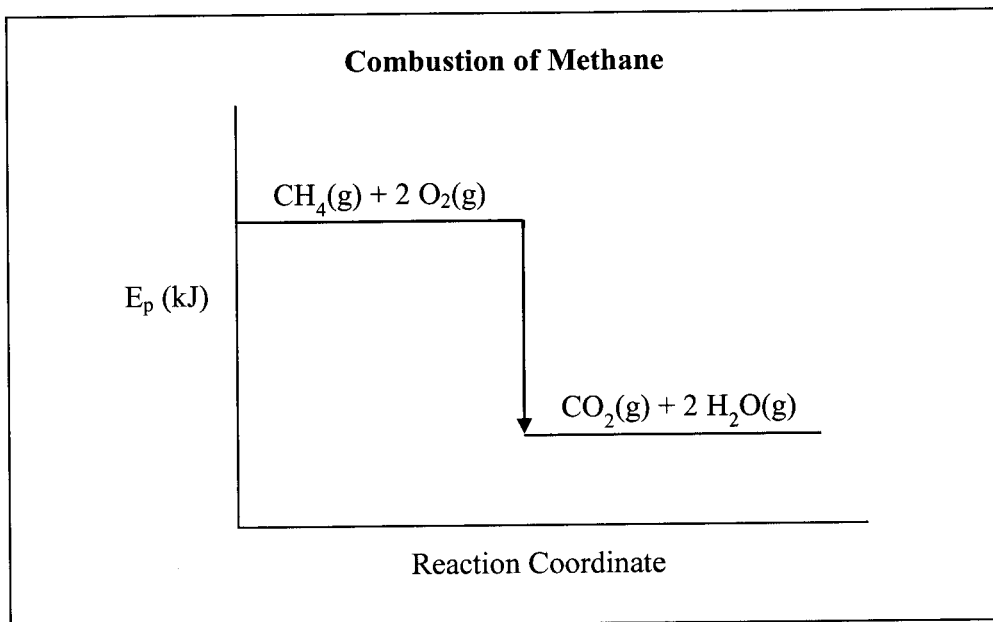


29. In the reaction represented by the equation above, energy is i the surroundings because the bonds in the products contain ii energy than the reactants.

The statement above is completed by the information in row

Row	i	ii
A.	absorbed from	more potential
B.	absorbed from	more kinetic
C.	released to	less potential
D.	released to	less kinetic

Use the following information to answer the next question.

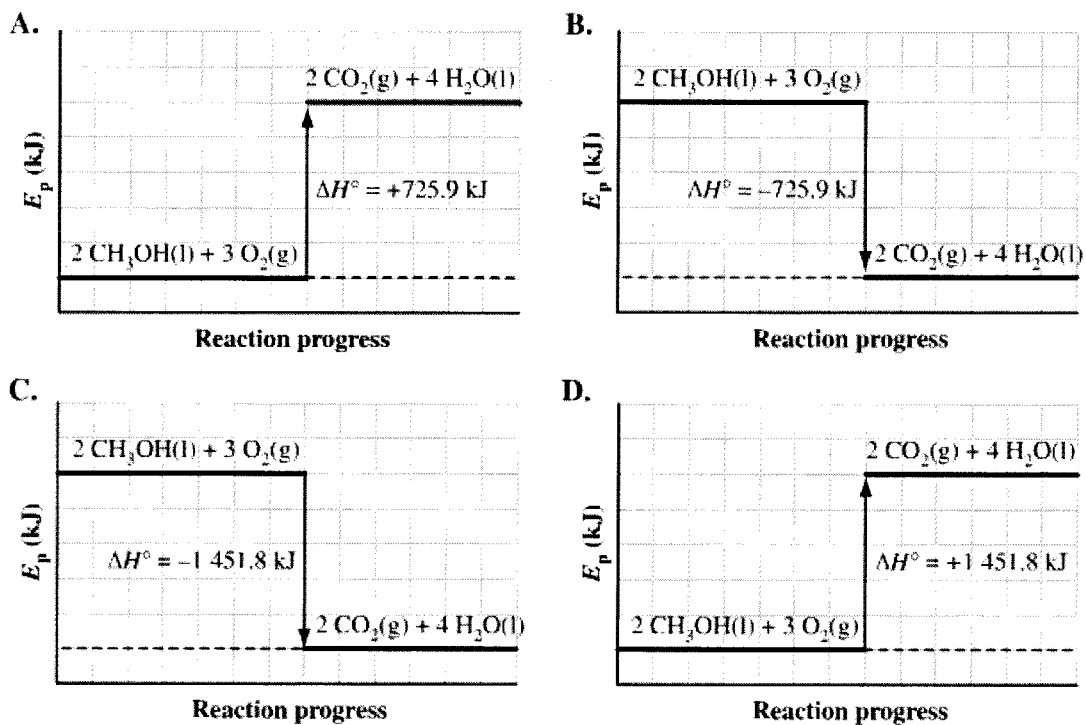


30. According to the diagram above, the enthalpy change for the combustion of methane gas is
- A. endothermic and represented by a negative ΔH value
 - B. endothermic and represented by a positive ΔH value
 - C. exothermic and represented by a negative ΔH value
 - D. exothermic and represented by a positive ΔH value

Use the following information to answer the next question.

Methanol is used as fuel in racing cars. The energy content of methanol can be determined by calorimetry in a closed system.

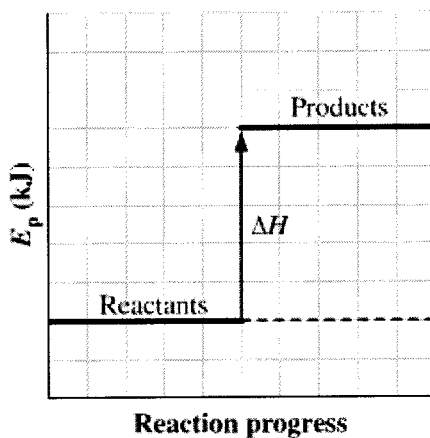
31. Which of the following diagrams represents the combustion of methanol in a closed system?



Use the following information to answer the next question.

Reactions

- | | | |
|---|---|--|
| 1 | Formation of carbon monoxide | $\Delta H^\circ = -110.5 \text{ kJ/mol}$ |
| 2 | Formation of nitrogen monoxide | $\Delta H^\circ = +91.3 \text{ kJ/mol}$ |
| 3 | Decomposition of hydrogen peroxide | $\Delta H^\circ = -187.8 \text{ kJ/mol}$ |
| 4 | Decomposition of dinitrogen tetroxide | $\Delta H^\circ = +11.1 \text{ kJ/mol}$ |
| 5 | $\text{Mn(s)} + \text{O}_2\text{(g)} \rightarrow \text{MnO}_2\text{(s)} + 520 \text{ kJ}$ | |
| 6 | $2 \text{ Ga(s)} + \text{O}_2\text{(g)} + 559 \text{ kJ} \rightarrow 2 \text{ GaO(g)}$ | |
| 7 | $\text{F}_2\text{(g)} + \text{O}_2\text{(g)} \rightarrow \text{F}_2\text{O}_2\text{(g)}$ | $\Delta H^\circ = +19.2 \text{ kJ/mol}$ |
| 8 | $\text{N}_2\text{O}_5\text{(g)} \rightarrow \text{N}_2\text{(g)} + \frac{5}{2} \text{ O}_2\text{(g)}$ | $\Delta H^\circ = -13.3 \text{ kJ/mol}$ |



Numerical Response

32. The enthalpy diagram above could be used to represent the reactions numbered _____, _____, _____, and _____.

Use the following information to answer the next question.

Chemicals	
1 $\text{O}_2(\text{g})$	4 $\text{H}_2\text{O}(\text{l})$
2 $\text{CO}(\text{g})$	5 $\text{H}_2\text{O}(\text{g})$
3 $\text{CO}_2(\text{g})$	6 $\text{C}_6\text{H}_{12}\text{O}_6(\text{aq})$

Numerical Response

33. Match the chemicals numbered above with the statements given below.

The reactants in photosynthesis are:

_____ and _____.
Record as the first digit Record as the second digit

The products of complete hydrocarbon combustion in an open system are:

_____ and _____.
Record as the third digit Record as the fourth digit

34. During photosynthesis, energy that originated from the i is converted to ii energy and stored in chemical bonds.

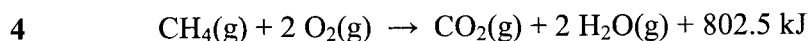
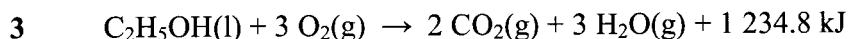
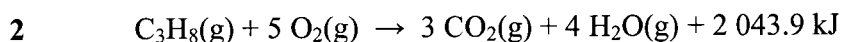
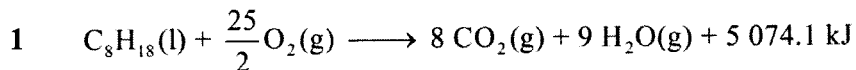
The statement above is completed by the information in row

Row	i	ii
A.	Sun	light
B.	Sun	chemical
C.	plant	light
D.	plant	chemical

35. Which of the following substances is produced during the combustion of octane in automobiles and is also produced during cellular respiration?
- A. $\text{NO}_2(\text{g})$
B. $\text{SO}_2(\text{g})$
C. $\text{CO}_2(\text{g})$
D. $\text{O}_2(\text{g})$

Use the following information to answer the next **two** questions.

Equations



36. The original source of energy for the reactants in the four reactions represented by the equations above is

- A. the Sun
- B. a fossil fuel
- C. a combustion reaction
- D. a decomposition reaction

Numerical Response

37. When ranked in order from the reaction that has the **smallest** enthalpy change per mole of carbon dioxide gas to the reaction that has the **largest** enthalpy change per mole of carbon dioxide gas, the reactions are _____, _____, _____, and _____.
- Smallest Largest

38. The products of photosynthesis are i , and the products of hydrocarbon combustion are ii .

The statement above is completed by the information in row

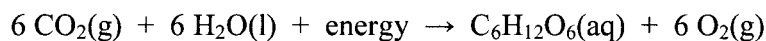
Row	i	ii
A.	$\text{CO}_2(\text{g})$ and $\text{H}_2\text{O}(\text{l})$	$\text{CO}_2(\text{g})$ and $\text{H}_2\text{O}(\text{g})$
B.	$\text{CO}_2(\text{g})$ and $\text{H}_2\text{O}(\text{l})$	$\text{CO}_2(\text{g})$ and $\text{H}_2\text{O}(\text{l})$
C.	$\text{C}_6\text{H}_{12}\text{O}_6(\text{aq})$ and $\text{O}_2(\text{g})$	$\text{CO}_2(\text{g})$ and $\text{H}_2\text{O}(\text{g})$
D.	$\text{C}_6\text{H}_{12}\text{O}_6(\text{aq})$ and $\text{O}_2(\text{g})$	$\text{CO}_2(\text{g})$ and $\text{H}_2\text{O}(\text{l})$

39. When the combustion of fossil fuels, such as propane, is compared to cellular respiration,
- A. both produce $\text{CO}_2(\text{g})$
 - B. respiration requires sunlight and combustion does not
 - C. combustion is endothermic and respiration is exothermic
 - D. combustion is exothermic and respiration is endothermic

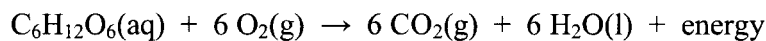
Use the following information to answer the next question.

The reactions for photosynthesis and cellular respiration are represented by the following equations.

Photosynthesis



Cellular Respiration



40. Which of the following statements accurately describes photosynthesis and cellular respiration?
- A. The products of photosynthesis have less potential energy than the reactants, whereas in cellular respiration, the reactants have less potential energy than the products.
 - B. Photosynthesis traps energy in the form of glucose, and cellular respiration absorbs energy required by all organisms.
 - C. Photosynthesis is an exothermic process, and cellular respiration is an endothermic process.
 - D. Photosynthesis requires energy, and cellular respiration produces energy.

Use the following information to answer the next question.

Silicon Halide	Molar Enthalpy of Formation
SiF ₄ (g)	-1 617 kJ/mol
SiCl ₄ (l)	-688 kJ/mol
SiBr ₄ (l)	-458 kJ/mol
SiI ₄ (s)	-190 kJ/mol

41. From the above list, the silicon halide that is the most thermally stable is

- A. SiF₄(g)
- B. SiCl₄(l)
- C. SiBr₄(l)
- D. SiI₄(s)

Use the following information to answer the next question.

I	$3 \text{ B(s)} + \text{X}_2\text{(g)} + 31.1 \text{ kJ} \rightarrow \text{B}_3\text{X}_2\text{(s)}$	
II	$\text{A}_2\text{(g)} + 4 \text{ B(s)} \rightarrow 2 \text{ AB}_2\text{(s)} + 42.0 \text{ kJ}$	
III	$\text{X}_2\text{(g)} + \text{Y}_2\text{(g)} \rightarrow 2 \text{ XY(g)}$	$\Delta H = -40.9 \text{ kJ}$
IV	$\text{C}_4\text{(s)} + 5 \text{ G}_2\text{(g)} \rightarrow 2 \text{ C}_2\text{G}_5\text{(s)}$	$\Delta H = +22.6 \text{ kJ}$

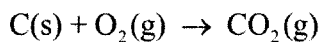
42. Which of the above reaction(s) is/are exothermic?

- A. II only
- B. III only
- C. I and IV only
- D. II and III

43. Which of the following statements is TRUE for an endothermic reaction?

- A. Energy is a product in the chemical equation.
- B. The temperature of the surroundings increases.
- C. The enthalpy change for the reaction is negative.
- D. The potential energy of the products is greater than the potential energy of the reactants.

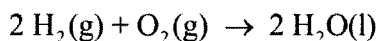
Use the following information to answer the next question.



$$\Delta H^\circ = -393.5 \text{ kJ}$$



$$\Delta H^\circ = -1959.2 \text{ kJ}$$



$$\Delta H^\circ = -571.6 \text{ kJ}$$

Numerical Response

44. The molar enthalpy of formation of cyclopropane, $3 \text{C(s)} + 3 \text{H}_2\text{(g)} \rightarrow \text{C}_3\text{H}_6\text{(g)}$, is $\pm$ _____ kJ/mol.

Use the following information to answer the next question.

1 methane

5 ethyne

2 ethene

6 nitrogen dioxide

3 benzene

7 butane

4 methanol

8 octane

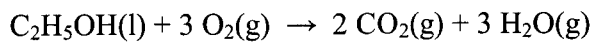
Numerical Response

45. The compounds that require energy during the formation from their constituent elements, listed in numerical order are _____, _____, _____, _____.

46. The molar enthalpy of combustion of propane in an open system is

- A. -103.8 kJ/mol
- B. -2043.9 kJ/mol
- C. -2219.9 kJ/mol
- D. -2251.5 kJ/mol

Use the following information to answer the next **two** questions.



47. During the reaction represented above, energy is i the surroundings because the reaction is ii .

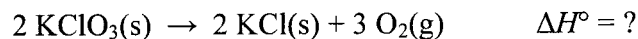
The statement above is completed by the information in row

Row	i	ii
A.	released to	exothermic
B.	released to	endothermic
C.	absorbed from	exothermic
D.	absorbed from	endothermic

Numerical Response

48. When 1.50 g of ethanol is burned, the energy released is _____ kJ.

Use the following information to answer the next question.

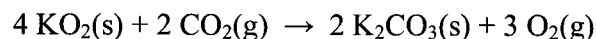


Numerical Response

49. The enthalpy change for the reaction represented by the equation above is +/- _____ kJ.

Use the following information to answer the next question.

Scuba divers who want to observe sharks will often use a rebreather, which is a self-contained breathing device. A rebreather allows divers to spend a longer time under water and prevents the release of exhaled bubbles that could scare away sharks. Potassium superoxide, $\text{KO}_2(\text{s})$, is used in some rebreathers to remove carbon dioxide from exhaled air. The equation for this reaction is shown below.



$$\Delta_f H^\circ_{\text{KO}_2} = -284.9 \text{ kJ/mol}$$

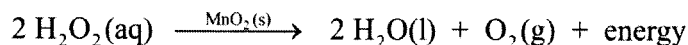
$$\Delta_f H^\circ_{\text{K}_2\text{CO}_3} = -1152.7 \text{ kJ/mol}$$

50. Which of the following rows identifies the type of reaction represented above and the type of ΔH value?

Row	Type of Reaction	ΔH Value
A.	Endothermic	Negative
B.	Endothermic	Positive
C.	Exothermic	Negative
D.	Exothermic	Positive

Use the following information to answer the next question.

Hydrogen peroxide solutions are commercially available because hydrogen peroxide decomposes very slowly. When $\text{MnO}_2(\text{s})$ is added to a solution of hydrogen peroxide, the hydrogen peroxide decomposes rapidly, as represented by the following equation.

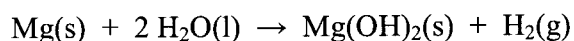


51. In the reaction represented by the equation above, $\text{MnO}_2(\text{s})$ acts to
- A. increase the activation energy
 - B. increase the energy released by the reaction
 - C. reduce the net energy released by the reaction
 - D. provide an alternative energy pathway for the reaction

Use the following information to answer the next question.

Some foods are available in insulated packages containing chemicals that produce an exothermic reaction that heats the food. When excess water and powdered magnesium metal react in the presence of an iron catalyst, energy is released. The reaction is represented by the following equation.

Reaction Used to Heat Packaged Food

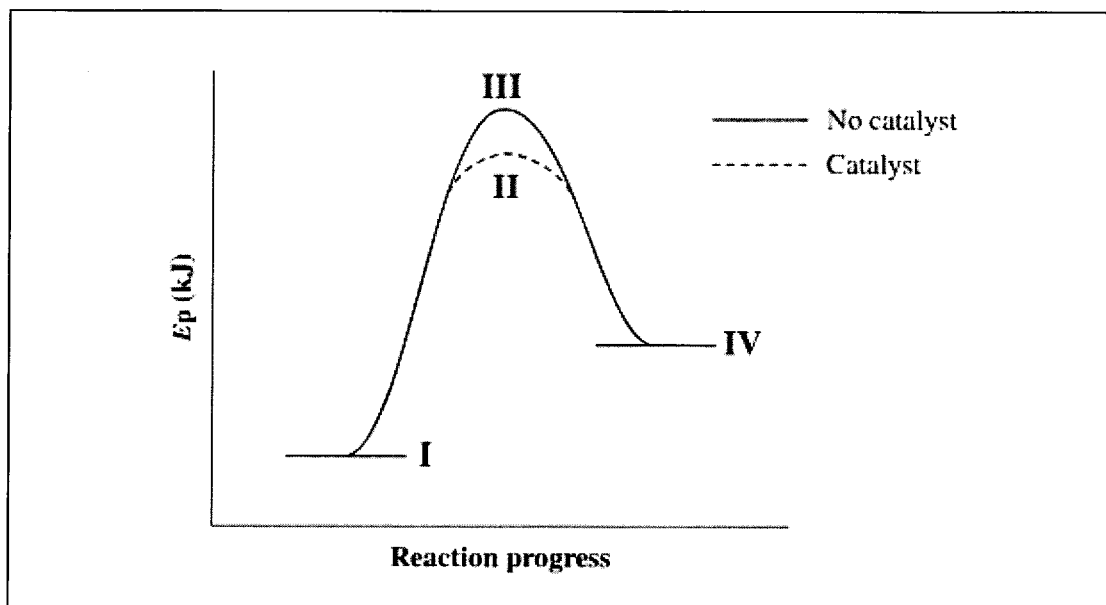


Statements About the Heating Reaction

- 1 The rate of reaction increases.
- 2 The ΔH° value for the reaction increases.
- 3 The ΔH° value for the reaction stays the same.
- 4 The potential energy of the products increases.

52. Which of the statements numbered above describes the effect of the iron catalyst on the reaction used to heat packaged food?
- A. 1 only
B. 1 and 3
C. 1, 2, and 4
D. 2 and 4 only

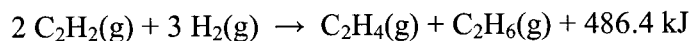
Use the following information to answer the next question.



53. The activation energy for the forward, catalyzed reaction is
- A. II minus I
 - B. III minus I
 - C. IV minus II
 - D. IV minus III
54. In the beer-making process, starch in the grain must be converted to sugar before fermentation can occur. The starch can be converted to sugar either by heating it or by adding a biological catalyst that
- A. allows the process to occur at higher temperatures
 - B. increase the amount of energy required for the process
 - C. increases the time it takes to convert the starch to sugar
 - D. decreases the time it takes to convert the starch to sugar

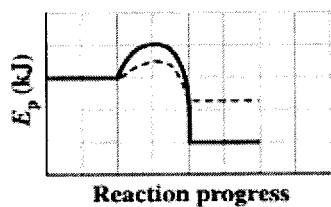
Use the following information to answer the next question.

One of the byproducts of the cracking process used at oil-refining plants is ethyne, $\text{C}_2\text{H}_2(\text{g})$. In the presence of a palladium catalyst, the ethyne forms ethane and ethane, as represented by the following equation.

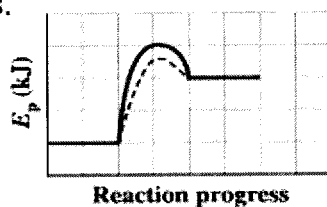


55. Which of the following energy pathway diagrams represents the energy changes for both the catalyzed and uncatalyzed reactions?

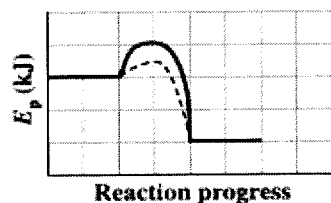
A.



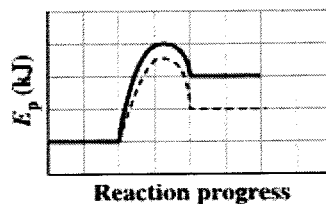
B.



C.



D.



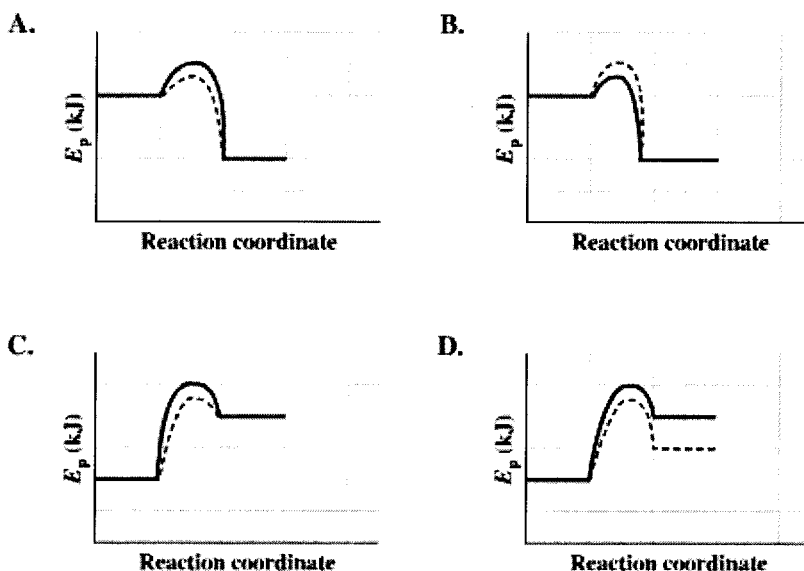
----- Catalyzed

————— Uncatalyzed

Use the following information to answer the next question.

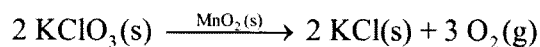
Dr. Richard Trotter has developed what could be the first cost-effective process for limiting methane emissions from underground coal mines. In this process, methane and oxygen are reacted at 800 °C in the presence of a catalyst. The products of this process are carbon dioxide gas and liquid water.

56. Which of the following potential energy diagrams represents both the catalyzed (----) and uncatalyzed (—) reactions for this process?



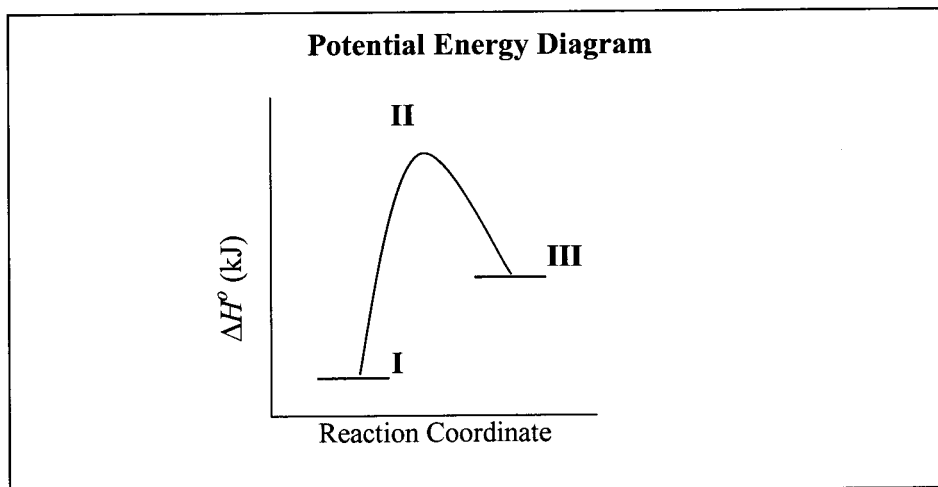
Use the following information to answer the next question.

The decomposition of hot potassium chlorate to solid potassium chloride and oxygen gas occurs at a faster rate in the presence of solid manganese(IV) oxide, as represented by the following equation.



57. When manganese(IV) oxide is used as a catalyst, the enthalpy change for the decomposition of potassium chlorate
- stays the same
 - becomes more exothermic
 - becomes more endothermic
 - changes from exothermic to endothermic

Use the following information to answer the next question.



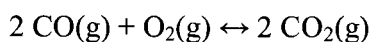
58. The energy barrier that must be overcome for the forward reaction to occur is called the i , and in the diagram above, it is the difference between ii .

The statement above is completed by the information in row

Row	i	ii
A.	enthalpy change	I and II
B.	enthalpy change	I and III
C.	activation energy	I and II
D.	activation energy	I and III

Use the following information to answer the next question.

Incomplete combustion in motor vehicles may lead to the formation of carbon monoxide gas, which is a health hazard in high concentrations. Carbon monoxide gas is converted to carbon dioxide gas in a catalytic converter before being emitted from the motor vehicle. This conversion is represented by the equation below.



59. The addition of a catalyst to the reaction represented by the equation above would i the energy transferred during the reaction and would ii the rate of the forward reaction.

The statement above is completed by the information in row

Row	i	ii
A.	increase	increase
B.	increase	not change
C.	not change	increase
D.	not change	not change

Use the following information to answer the next question.

Type of Reaction

1 Exothermic

2 Endothermic

Numerical Response

60. Match the type of reaction numbered above that corresponds to each of the terms listed below.

Photosynthesis	_____	(Record as the first digit)
Cellular Respiration	_____	(Record as the second digit)
Formation of Glucose	_____	(Record as the third digit)
Combustion of Propane	_____	(Record as the fourth digit)

61. *The energy changes that occur when propane undergoes combustion are primarily due to changes in i energy resulting from ii .*

The statement above is completed by the information in row

Row	<i>i</i>	<i>ii</i>
A.	potential	a rearrangement of bonds
B.	potential	an increase in molecular motion
C.	kinetic	a rearrangement of bonds
D.	kinetic	an increase in molecular motion

Unit A: Thermochemical Changes – Answers

- | | | |
|----------|----------|-------|
| 1. 2.81 | 31. C | 61. A |
| 2. 27.5 | 32. 2467 | |
| 3. C | 33. 3435 | |
| 4. C | 34. B | |
| 5. A | 35. C | |
| 6. B | 36. A | |
| 7. A | 37. 3124 | |
| 8. B | 38. C | |
| 9. A | 39. A | |
| 10. B | 40. D | |
| 11. D | 41. A | |
| 12. 30.9 | 42. D | |
| 13. D | 43. D | |
| 14. A | 44. 78.7 | |
| 15. A | 45. 2356 | |
| 16. D | 46. B | |
| 17. C | 47. A | |
| 18. 39.5 | 48. 40.2 | |
| 19. D | 49. 77.6 | |
| 20. C | 50. C | |
| 21. A | 51. D | |
| 22. 4.52 | 52. B | |
| 23. C | 53. A | |
| 24. A | 54. D | |
| 25. 41.1 | 55. C | |
| 26. C | 56. A | |
| 27. B | 57. A | |
| 28. B | 58. C | |
| 29. C | 59. C | |
| 30. C | 60. 2111 | |

